

Supplemental Material:

Three local interactions of the ten-dimensional self-dual five-form that no stress-tensor flow generates

AUTHOR LIST — HUMAN ACTION REQUIRED¹

¹*AFFILIATIONS — HUMAN ACTION REQUIRED*

The Letter stands on its own. Nothing in this Supplement is needed to believe the central theorem; everything here is detail a specialist may want to check. The one thing that cannot be moved here is the logic of the theorem itself, and it is not.

CONVENTIONS

Signature $(-, +, \dots, +)$; $\epsilon_{01\dots 9} = +1$; $(\star F)_I = \frac{1}{(d-p)!} \epsilon_I^J F_J$. For $d = 10$, $p = 5$, $\star^2 = (-1)^{25} \text{sgn det } \eta = +1$, computed at runtime rather than assumed. Self-duality is $\star F = +F$, projector $(1 + \star)/2$ with $1/2$ the modular inverse of 2. Clifford normalisation $\{\Gamma^\mu, \Gamma^\nu\} = 2\eta^{\mu\nu}$; positive chirality is $\Lambda^{\text{even}} W$; the antisymmetrised five-gamma carries unit weight on the identity permutation.

Every convention is pinned as a named constant in `sdbridge/conventions.py`; downstream code imports the constant and may not re-derive a sign locally.

ARITHMETIC AND PRIMES

All ranks are exact over $\mathbb{F}_p$. Fitting primes 32749, 32719; holdout primes 32717, 32713, 32707. Modular rank bounds the characteristic-zero rank from below unconditionally when the matrix is an integer reduction, which holds here because the coordinate basis is integral with entries in $\{-1, 0, +1\}$. No upper bound anywhere in this work is taken from a modular computation.

SAMPLE PROTOCOL

A registry of 78 self-dual five-forms: 10 sparse, 8 structured, 40 generic, 20 holdout drawn from a disjoint seed stream. Every sample is hashed, checked for exact self-duality, and its bridge image checked gamma-traceless before use. Change-of-basis maps are fitted on the non-holdout samples and validated on the holdout samples only.

THE BRIDGE

$S_{ab} = \frac{1}{5!} F_{\mu_1\dots\mu_5} (\Gamma^{\mu_1\dots\mu_5})_{ab}$. Verified at each prime: kernel equals the anti-self-dual 126 and image equals the gamma-traceless 126, both by two-way span containment; forward rank 126, not 136; the constructed left inverse composes with the forward map to give the self-dual projector exactly.

The oscillator frame has real signature $(5, 5)$, computed from all one hundred anticommutators of the frame's own operators. $(5, 5)$ and $(1, 9)$ are inequivalent real forms; over $\mathbb{F}_p$ they are congruent because both metrics have discriminant -1 up to squares, and the transition matrix is constructed explicitly rather than assumed to exist.

The spinor index placement in σ_{ab}^μ is undocumented upstream. It was *determined* by requiring exact equivariance: of eight candidate placements exactly one reproduces the transformed image on every component with a single scalar. This is a reconstruction of a convention, not a citation of one.

DEGREE-8 SPAN EQUALITY

At four primes, trace rank 7, spinor rank 7, union rank 7, with two-way containment and holdout-validated change of basis. Dropping each spinor family in turn and re-ranking: port graphs 7, products 7, structured Ω/Θ 7, **tensor words** 6. Only the tensor words are indispensable, which locates the direction a port-graph-only family misses.

EXACT JACOBIAN

Analytic derivatives throughout; no finite differences. Port graphs are differentiated by amputation, tensor words by the product rule on the trace, and the Ω/Θ family by exact modular interpolation cross-checked against an independent polarisation. Euler homogeneity $\sum_r c_r \partial_r I = (\deg I) I$ holds for every evaluated candidate.

Rank 81; per-degree block ranks 1, 2, 6, 12, 61; cumulative 1, 3, 9, 21, 81. No row normalisation is used, and over $\mathbb{F}_p$ none would be meaningful. Of 83 scheduled candidates, 82 carry terminal status **evaluated**; one exceeded the wall-clock budget and is recorded as **evaluation_error** rather than dropped. It cannot change a rank already at the analytic maximum.

WHAT IS COMPUTER-ASSISTED

The dimensions are ranks of explicitly recorded integer matrices. The repository ships the matrices, the pivot rows and columns, and a script that recomputes each rank from stored data without re-running any search, so the ranks are checkable without trusting the enumeration. The enumeration itself is the step that rests on search; it is corroborated by an independent formula-free reverse recovery and by the spinor computation, but it is not a symbolic proof.

SOFTWARE

Three packages with no shared implementation: the tensor implementation, the bridge, and an external third-party spinor archive that is analysed but never redistributed. Import audits in both directions are recorded in `docs/SOURCE_OVERLAP_AUDIT.md`. Implementation independence holds; clean-room independence is not claimed.